

Exercise Sheet for Maths Material Covered in Week 11

Matrix Multiplication, Identity and Inverse

(1) Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

(a) Find a 2×2 matrix B (if it exists) such that $BA = I$?

(b) Find a 2×2 matrix C (if it exists) such that $AC = I$?

Comment: After answering part (a), is there an easy way to answer part (b)?

(2) Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$ and $I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

(a) Find a 3×3 matrix B (if it exists) such that $BA = I$?

(b) Find a 3×3 matrix C (if it exists) such that $AC = I$?

(3) Let $X = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 4 \\ 1 & 0 & 5 \end{pmatrix}$ and $I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

(a) Find a 3×3 matrix Y (if it exists) such that $XY = I$?

(b) Find a 3×3 matrix Z (if it exists) such that $ZX = I$?

(4) Let $a, b, c, d \in \mathbb{Q}$ be such that none of them are zero. Let $X = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$. If $ad - bc = 0$, then show that there is no 2×2 matrix Y whose entries are rational numbers and which satisfies $XY = I$.